Radical Expressions and Rational Exponents — Finding the Illegal Split

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A radical distributes over a **product**, never over a **sum**: $\sqrt{ab} = \sqrt{a}\,\sqrt{b}$ is a rule, but $\sqrt{a+b}$ is *not* $\sqrt{a} + \sqrt{b}$.

- Several problems below show somebody's work. Find the step where a radical (or a 1/2 power) was distributed across a sum.
- Then do the problem correctly and put the correct value on the answer line. Show the testing or the algebra that settles it.

1. Test the claim with numbers you can check. Work out $\sqrt{9+16}$ by adding inside the radical first. Separately, work out $\sqrt{9} + \sqrt{16}$, which comes to 7. Write the correct value of $\sqrt{9+16}$ on the answer line.

Answer: ____________________

2. Find and fix. Priya writes

$$\sqrt{36+64} \;=\; \sqrt{36} + \sqrt{64} \;=\; 14.$$

Name the step that is not legal, then compute $\sqrt{36+64}$ correctly.

Answer: ____________________

3. Bo replaces $\sqrt{5+7}$ with $\sqrt{5}+\sqrt{7}$ and reports 4.88. That is the same illegal split, and this time neither number is a whole number, so it is easy to miss. Compute $\sqrt{5+7}$ rounded to two decimal places.

Answer: ________________

4. **Find and fix.** Dana writes $\sqrt{50}+\sqrt{18} = \sqrt{68} \approx 8.25$, pulling two separate radicals into one radical of the sum. Simplify $\sqrt{50}$ and $\sqrt{18}$ separately with the product rule, combine the like radicals, and give the exact answer together with its decimal value to two places.

Answer: ________________

5. The same claim written with rational exponents. Kofi starts from $(25+144)^{1/2}$, gives each term its own one-half power, and gets $25^{1/2} + 144^{1/2} = 17$. Compute $(25+144)^{1/2}$ correctly.

Answer: ________________

6. Find and fix. Solve $\sqrt{x+9} = 5$. Jamal writes $\sqrt{x} + \sqrt{9} = 5$, so $\sqrt{x} = 2$ and $x = 4$. Substitute $x = 4$ into the original equation to show Jamal's answer fails, then solve the equation correctly by squaring the whole left side at once.

Answer: ________________

7. A rectangular patio measures 20 feet by 21 feet. Its diagonal brace has length $\sqrt{20^2 + 21^2}$ feet. Sam writes the diagonal as $20 + 21 = 41$ feet. Find the exact length of the diagonal brace, in feet.

Answer: ________________ ft

8. Explain, then solve.
(a) Square the expression $\sqrt{a}+\sqrt{b}$ and compare the result with $a+b$. Use what you get to explain, in your own words, why $\sqrt{a+b}=\sqrt{a}+\sqrt{b}$ fails for almost every pair of numbers.
(b) Exactly one value of x with $x \geq 0$ makes $\sqrt{x+25}=\sqrt{x}+5$ a true statement. Find that value, and write it on the answer line.

Answer: ________________